

Test IV

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Trigonometry).

II) Show all work.

III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) You will need a calculator for this test. When pi is required, use the π button on your calculator. Do not use 3.14.

V) Each problem is worth four points for a total of 100 points.

Find the value of the expression using a calculator.
Round to the nearest thousandth.

① $\sin\left(\frac{\pi}{8}\right)$

② Find the area of a sector of a circle with radius 5 meters and central angle $3\pi/10$. Round to the nearest tenth.

③ i) Convert the following angle measure to radian units.
 300°

ii) Convert the following angle measure to degree units.
 $\frac{5\pi}{2}$

④ Find the values of the following expressions without a calculator.

i) $\sin(2\pi)$

ii) $\sec\left(\frac{2\pi}{3}\right)$

⑤ Find the value of the following expression without a calculator.

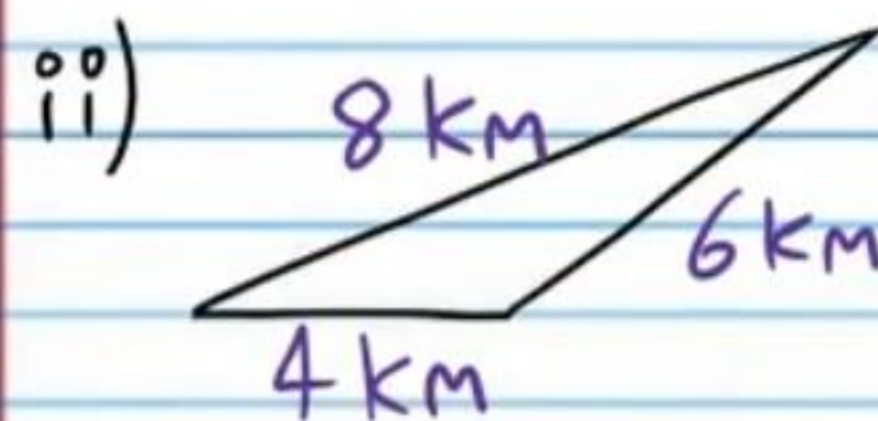
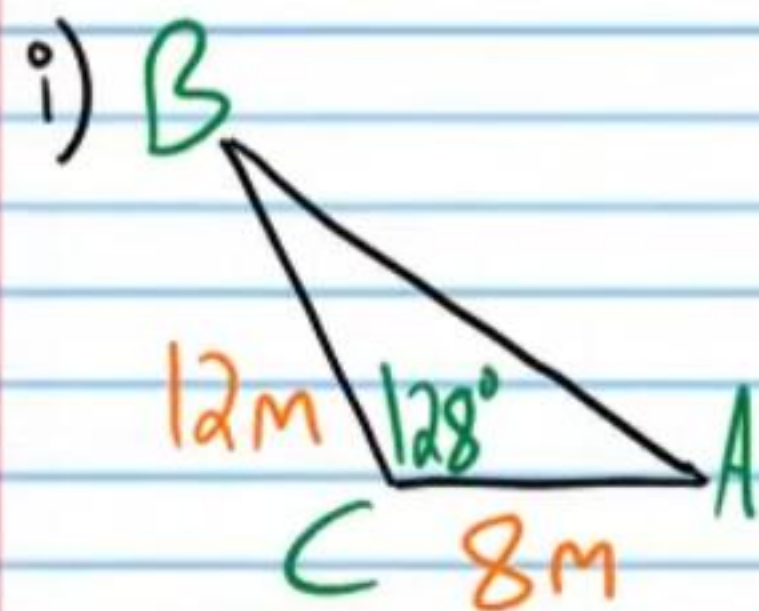
$\operatorname{arccsc}\left(-\frac{2}{\sqrt{3}}\right)$

⑥ Find the values of the following expressions with a calculator. Round to the nearest ten thousandth.

i) $\sec 141^\circ$

ii) $\csc^{-1}(4)$

⑦ Find the area of each triangle below. Round to the nearest hundredth.



⑧ Solve if $0 < \theta < 2\pi$. Do not use a calculator. Write your answer in radian units.

$$\sin \theta = \frac{3}{2\sqrt{3}}$$

⑨ Find β if $0 \leq \beta \leq 2\pi$. Write your answer in radian units. Do not use a calculator.

$$3\cos\beta - 1 = 5\cos\beta$$

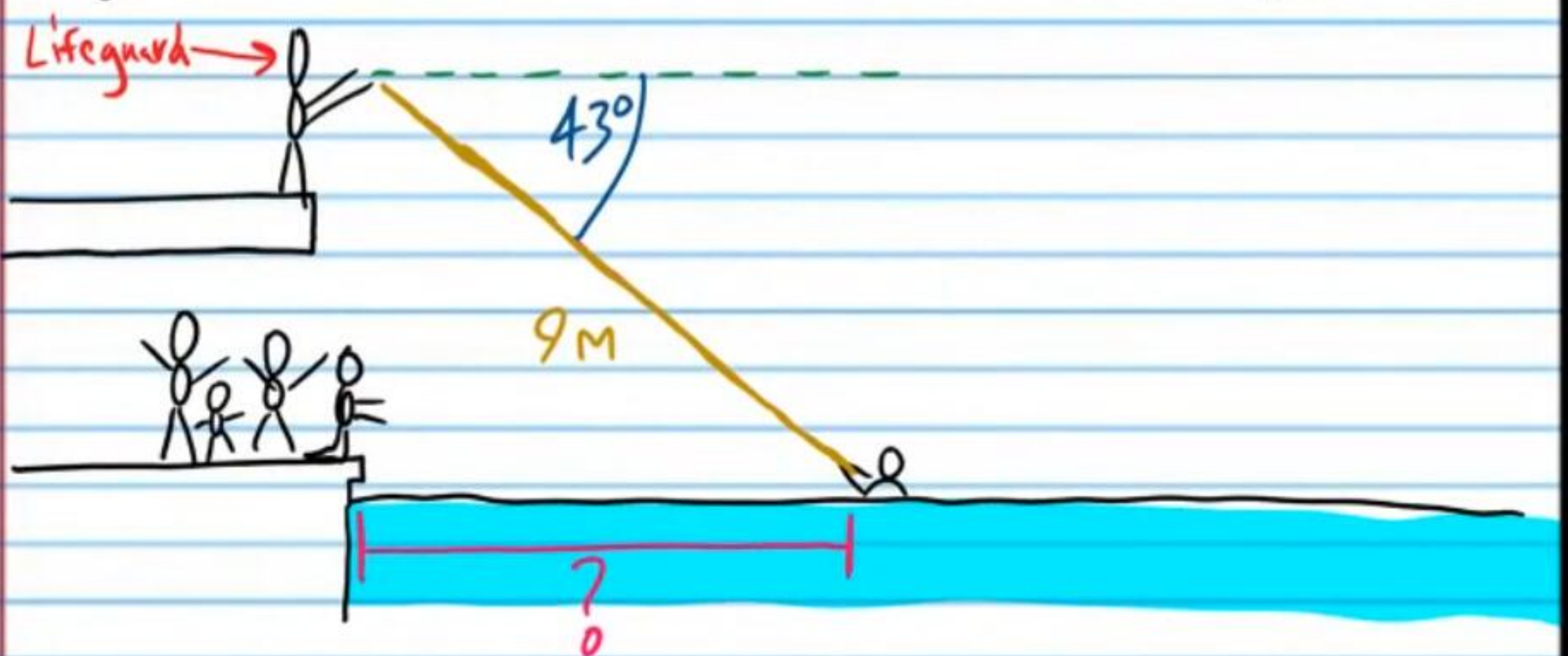
⑩ Solve if $0 \leq w < 360^\circ$. Round to the nearest hundredth.

$$-2\cot w + 5 - 4\cot w = 9 + \cot w$$

⑪ Write an equivalent expression without trigonometric notation. Assume that x is positive.

$$\cos(\tan^{-1} \frac{1}{x})$$

(12) A lifeguard rescues someone from a pool with a rope that is 9 meters long. If the angle of depression of the rope is 43° , what is the horizontal distance from the man in the pool to people at the edge of the pool that will pull him out?



Find the value of each expression *without a calculator*.

(13) $\sin(\csc^{-1} 2)$

$$\textcircled{14} \cot(\cos^{-1}(0))$$

$$\textcircled{15} \sec(\sin^{-1}(-\frac{1}{\sqrt{2}}))$$

⑩ $\cot(\cot^{-1}1)$

Find the value of each expression *without a calculator.*

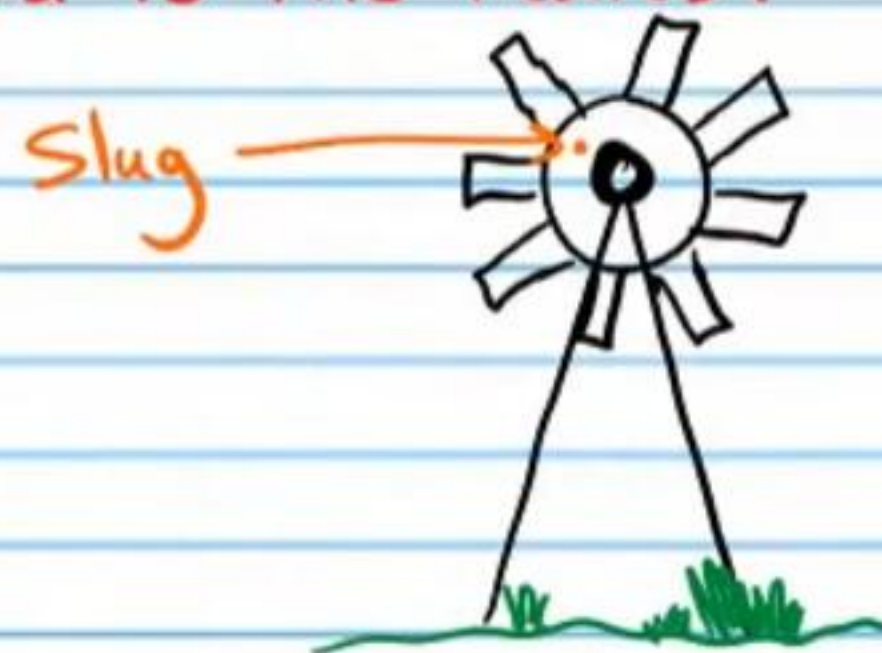
⑪ $\cos^{-1}(\sin 330^\circ)$

(18) i) $\sec^{-1}(\sec 2\pi)$

ii) $\csc^{-1}(\csc \frac{\pi}{6})$

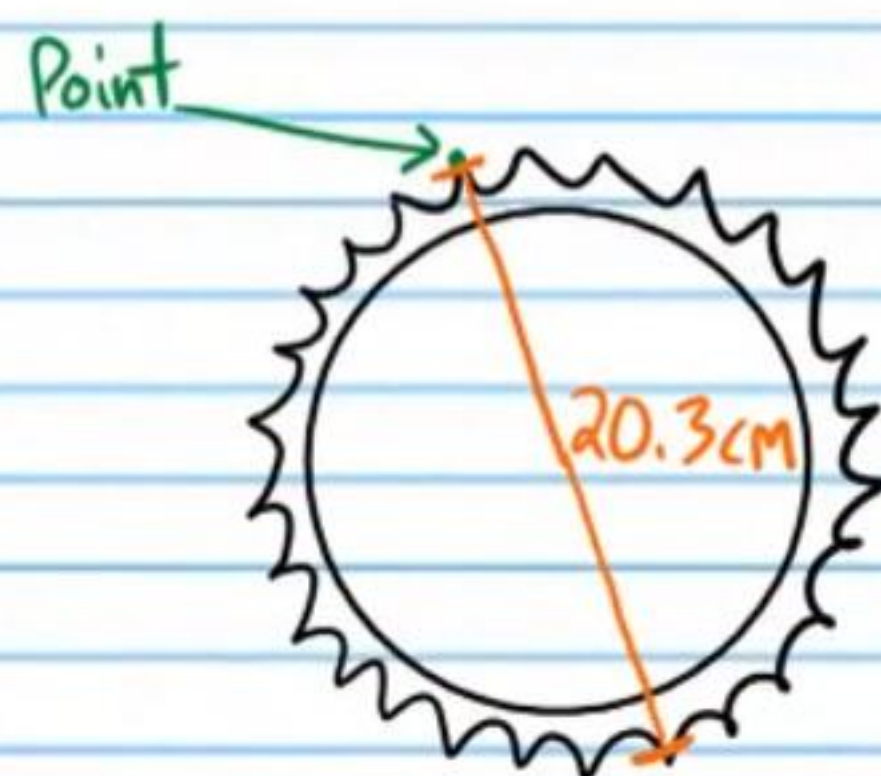
(19) $\cot^{-1}(\tan(-\frac{5\pi}{4}))$

(20) A windmill completes one rotation in 1.61 seconds. How many inches per second does a slug stuck to the mill move if it is 15 inches from the center of the mill? Round to the nearest tenth.

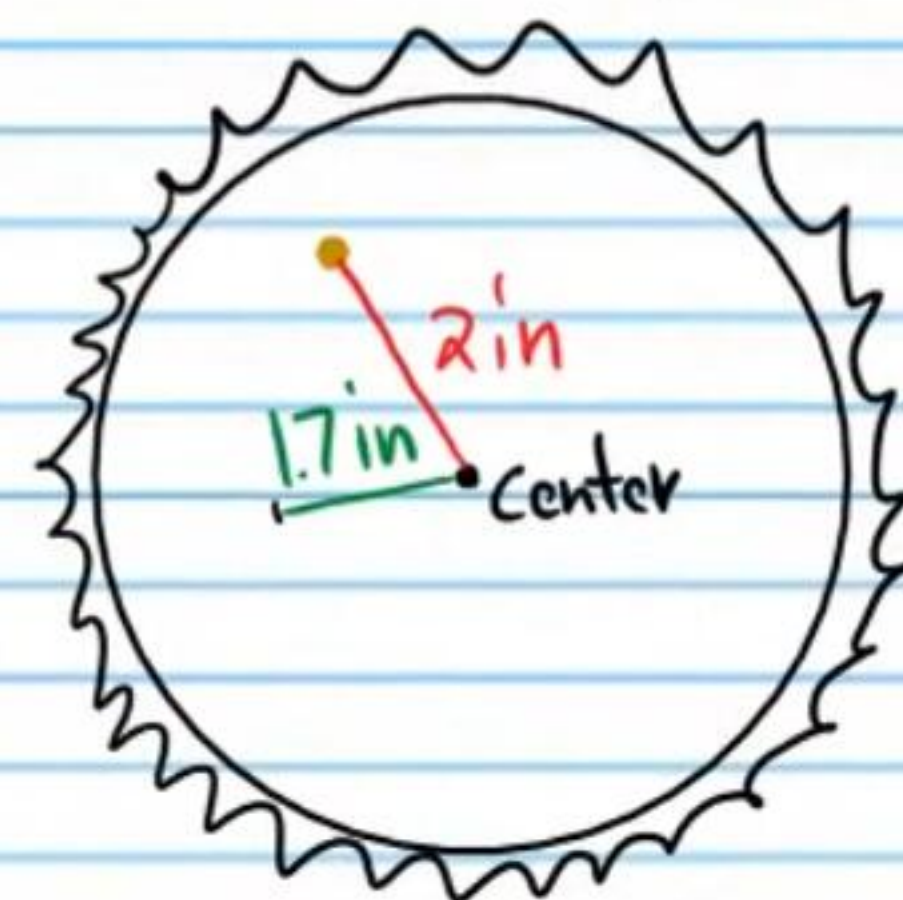


(21) A Ferris wheel makes three complete rotations in 37 seconds. If a child is standing 12 feet from the center of the wheel, what is her linear velocity in feet per minute? Round to the nearest whole number.

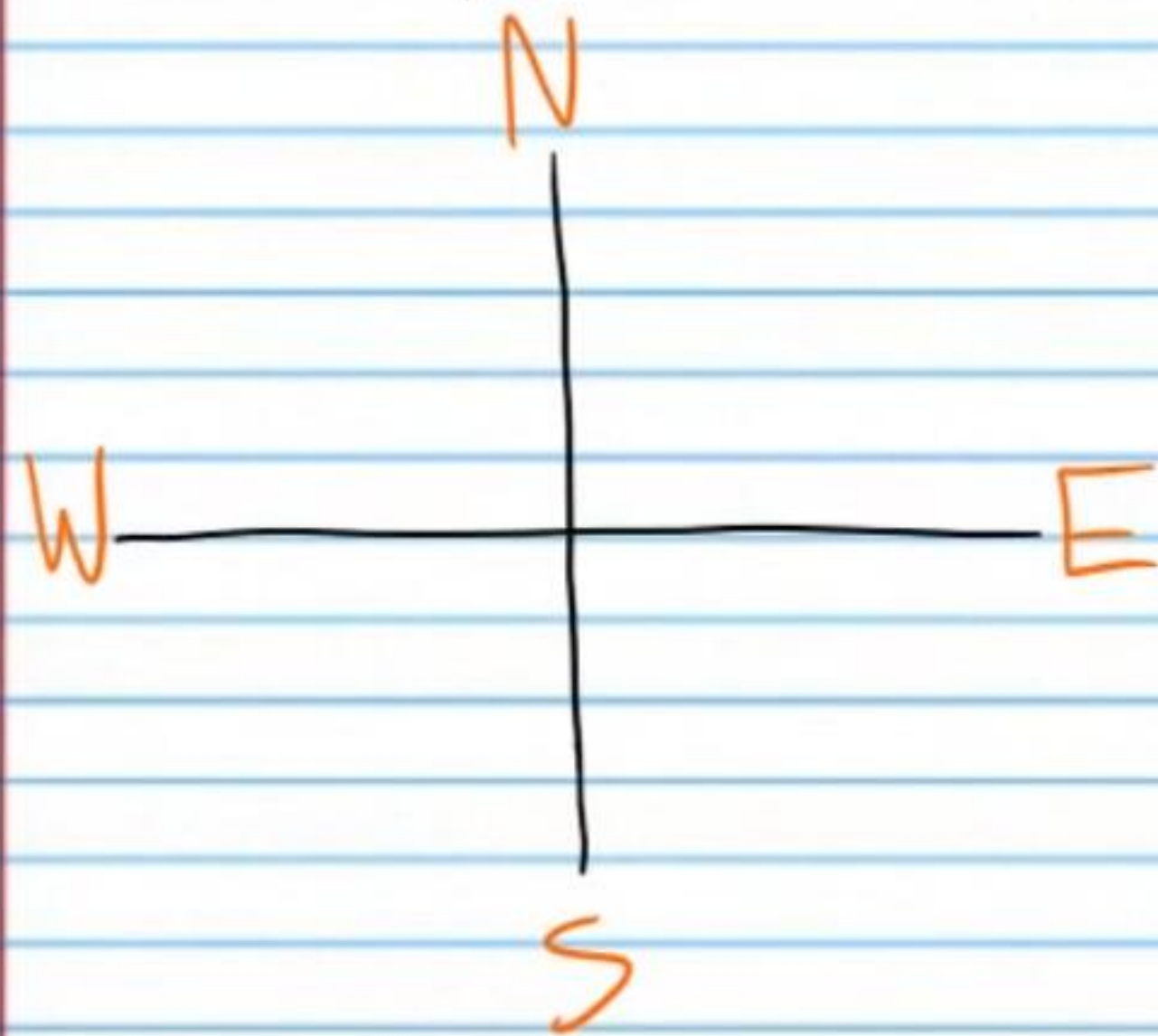
② A sprocket rotates at 8,000 rpm. If the diameter of the sprocket is 20.3 centimeters, how fast does a point on the edge of a tooth move in centimeters per second? Round to the nearest hundredth.



③ A speck of mud is on a saw blade 2 inches from the center and has a linear velocity of 50 inches per second. What is the angular velocity of the blade? What is the linear velocity of a small scratch 1.7 inches from the center in inches per second?

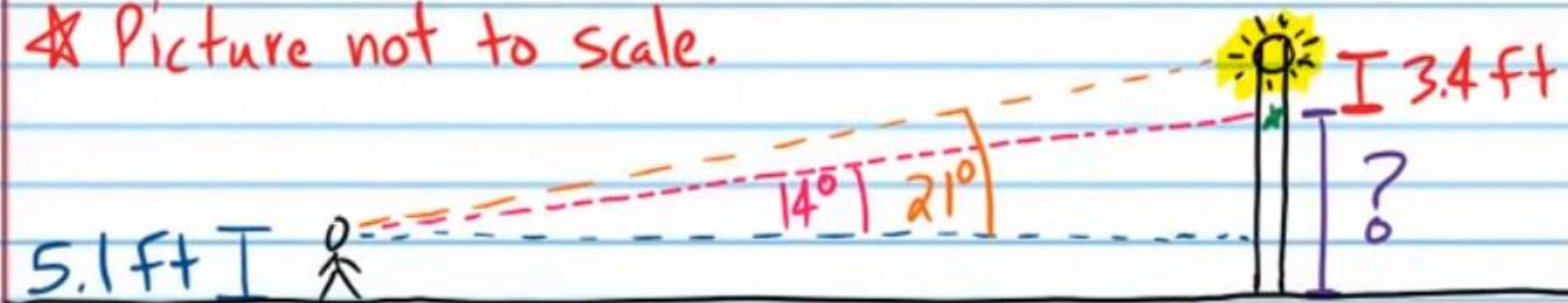


②4 A gps device (global positioning system) shows that a bicyclist must ride 40 miles on a bearing 30 degrees north of east ($E 30^\circ N$) to get to Melbourne Bridge. The device also shows that the bicyclist must ride 47 miles on a bearing 20 degrees south of east ($E 20^\circ S$) to get to his friend Bill's house. How far apart are Melbourne Bridge and Bill's house?



②5 A man looks up at a streetlight with a 21° angle of elevation. He then looks at a lizard on the light pole with a 14° angle of elevation. The light and the lizard are 3.4 feet apart. If the man is 5.1 feet tall, how far is the lizard from the ground?

★ Picture not to scale.



Test IV Answers

① 0.383

② 11.8 m^2

③ i) $\frac{5\pi}{3}$

ii) 450°

④ i) 0

ii) -2

⑤ -60°

⑥ i) -1.2868

ii) 14.4775°

⑦ i) 37.82 m^2

ii) 11.62 km^2

⑧ $\frac{\pi}{3}, \frac{2\pi}{3}$

⑨ $\frac{2\pi}{3}, \frac{4\pi}{3}$

⑩ $119.74^\circ, 299.74^\circ$

⑪ $\frac{x}{\sqrt{x^2+1}}$

⑫ 6.58 m

⑬ $\frac{1}{2}$

⑭ 0

⑮ $\sqrt{2}$

⑯ 1

⑰ 120°

⑱ i) 0°

ii) $\frac{\pi}{6}$

⑲ 135°

⑳ 58.5 in/s

㉑ 367 ft/min

㉒ $8,503.24 \text{ cm/s}$

㉓ $\omega = 25 \text{ rad/s}, v = 42.5 \text{ in/s}$

㉔ 37.31 mi

㉕ 11.4 ft